



SHANGHAI, CHINA

ISCAS 2026

IEEE INTERNATIONAL SYMPOSIUM ON CIRCUITS AND SYSTEMS

**2026 IEEE International Symposium
on Circuits and Systems**

Circuits and Systems for Intelligent Society

Live Demonstration: An Agile FPGA-Overlaid CGRA SoC for High-Efficiency Computing

Jiahang Lou*, Jianrong Zhang*, Yuan Dai, Zewei Zhong, Huan Lin, Wenbo Yin and Lingli Wang#

State Key Laboratory of Integrated Chips and Systems, Fudan University, Shanghai, China

Email: jhlou22@m.fudan.edu.cn, zhangjr25@alumni.fudan.edu.cn, llwang@fudan.edu.cn



Poster NO.2672

INTRODUCTION

The growing computational demands of AI and high-performance computing architectures that balance efficiency, performance, and flexibility. While FPGAs offer reconfigurability, their fine-grained nature can introduce significant overhead for many applications. Coarse-Grained Reconfigurable Arrays (CGRAs) address this by operating at the word level, thereby reducing configuration overhead and improving energy efficiency while retaining flexibility. This demo illustrates a practical realization through an FPGA-based CGRA overlay. The system decouples the accelerator from a specific CPU, facilitating portable and scalable integration. We also highlight a co-design approach, where a fully integrated hardware SoC is paired with a dedicated software toolchain to enable agile development and deployment of applications, from high-level description to hardware execution.

FRAMEWORK

We proposed a full-stack CGRA SoC Implementation. The hardware features a parameterized CGRA, with an MLIR based compiler frontend[1]. A dedicated runtime library is also developed.

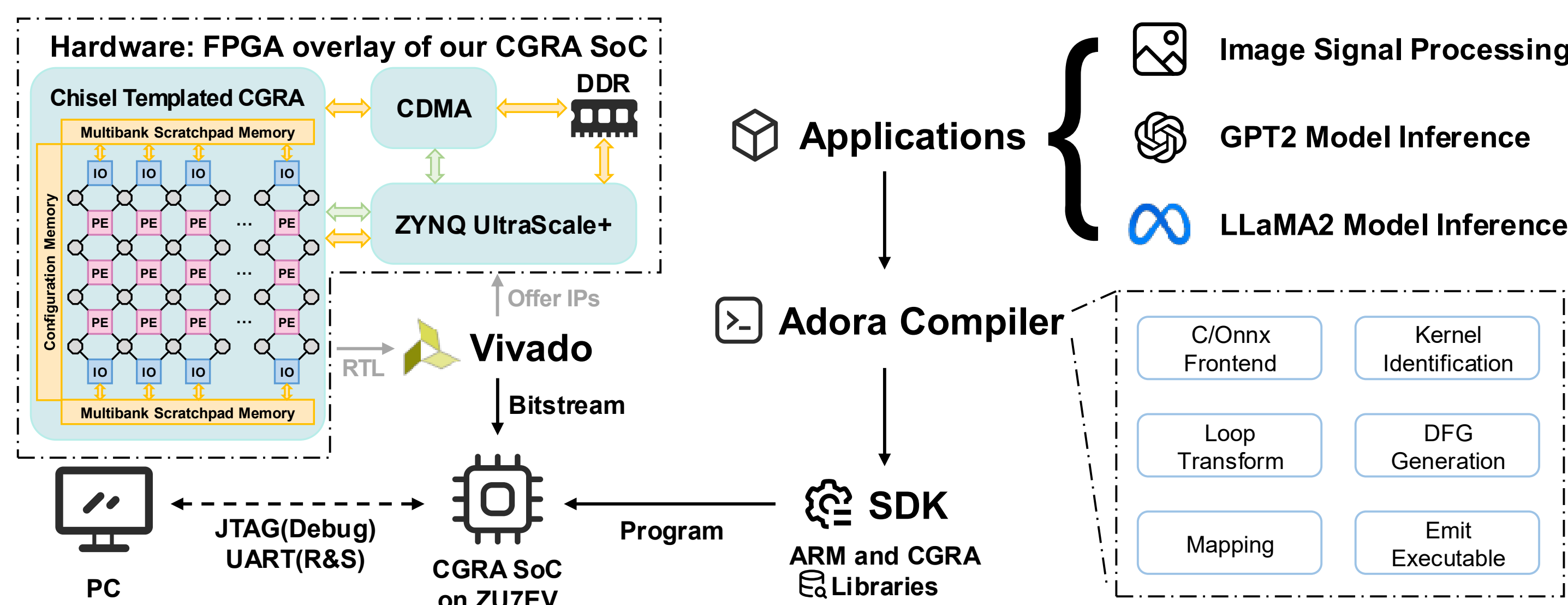


Fig.1. Full-Stack Framework Overview

EVALUATION RESULTS

In the LLaMA2-42M model inference task, our FPGA-Overlaid CGRA SoC achieves a **10.49×** speedup in kernel execution cycles and a **7.04×** speedup in overall execution cycles compared to the ARM Cortex-A53 CPU baseline. We also provide an ASIC synthesis result of the core array using the TSMC 40nm standard-cell library, which achieves a maximum operating frequency of **505MHz**.

TABLE 1: END TO END COMPARISON of LLM INFERENCE

	Kernel Cycle	Overall Cycle	Latency(ms)	Power(W)	Area
ARM Cortex-A53	2.195×10^8	2.236×10^8	168.12@1.33GHz	≈ 0.3	—
AuFORA CGRA SoC	$2.093 \times 10^7^*$ (Ratio:10.49 $\times$)	$3.174 \times 10^7^+$ (Ratio:7.04 $\times$)	456.42@70MHz (FPGA ZU7EV)	1.861	170688 LUTs 70115 FFs 224 DSPs 68 BRAMs
			66.48@500MHz (ASIC TSMC40nm)	0.371	3.652 mm ²

* Including data transfers, array configuration and execution.

† Including DMA requests, cache flushes and AXI-Lite controls.

DEMONSTRATION SETUP

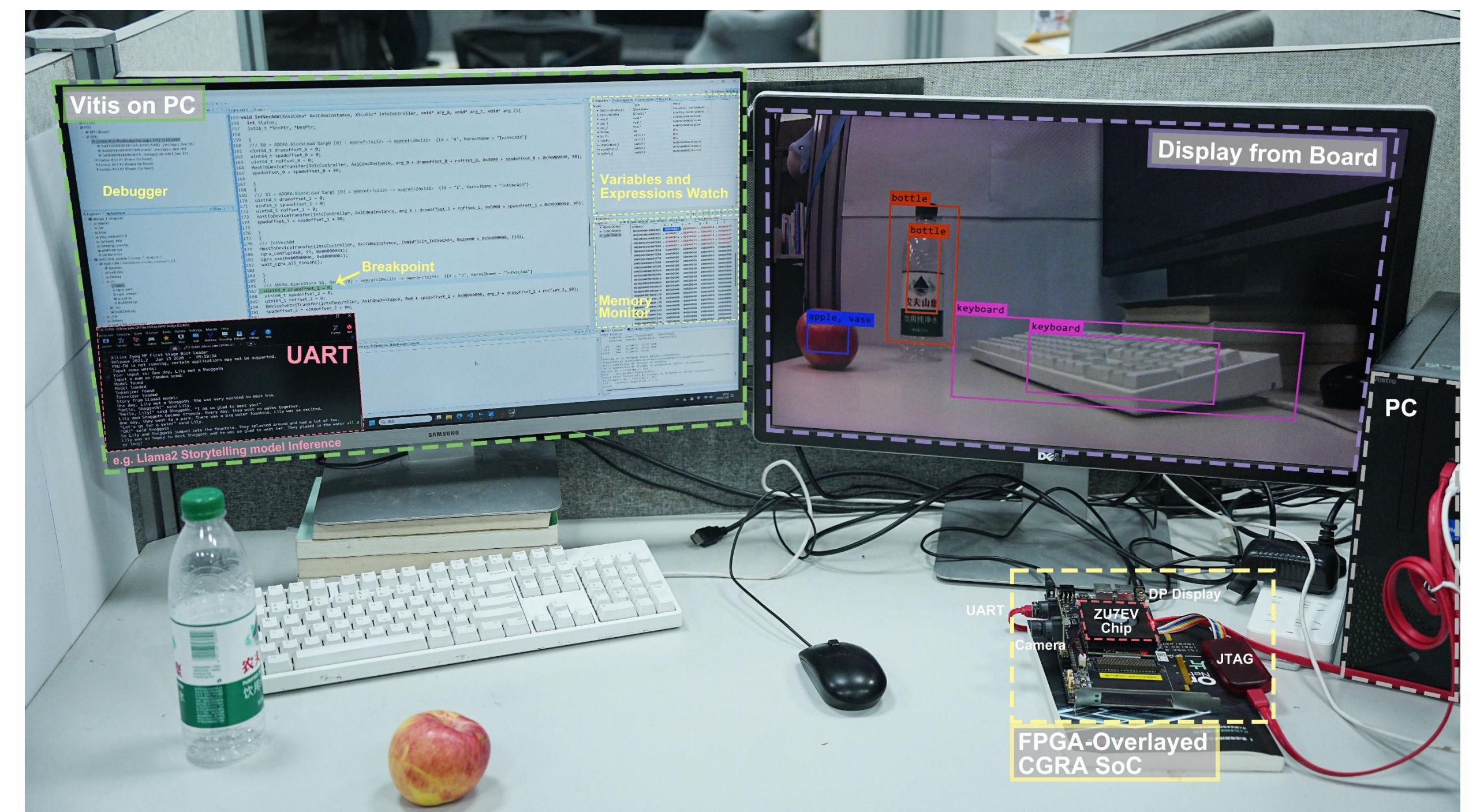


Fig.2. Hardware System Implementation and Debugger. Left: LLM inference through UART. Right: YOLOv3Tiny object detection from camera

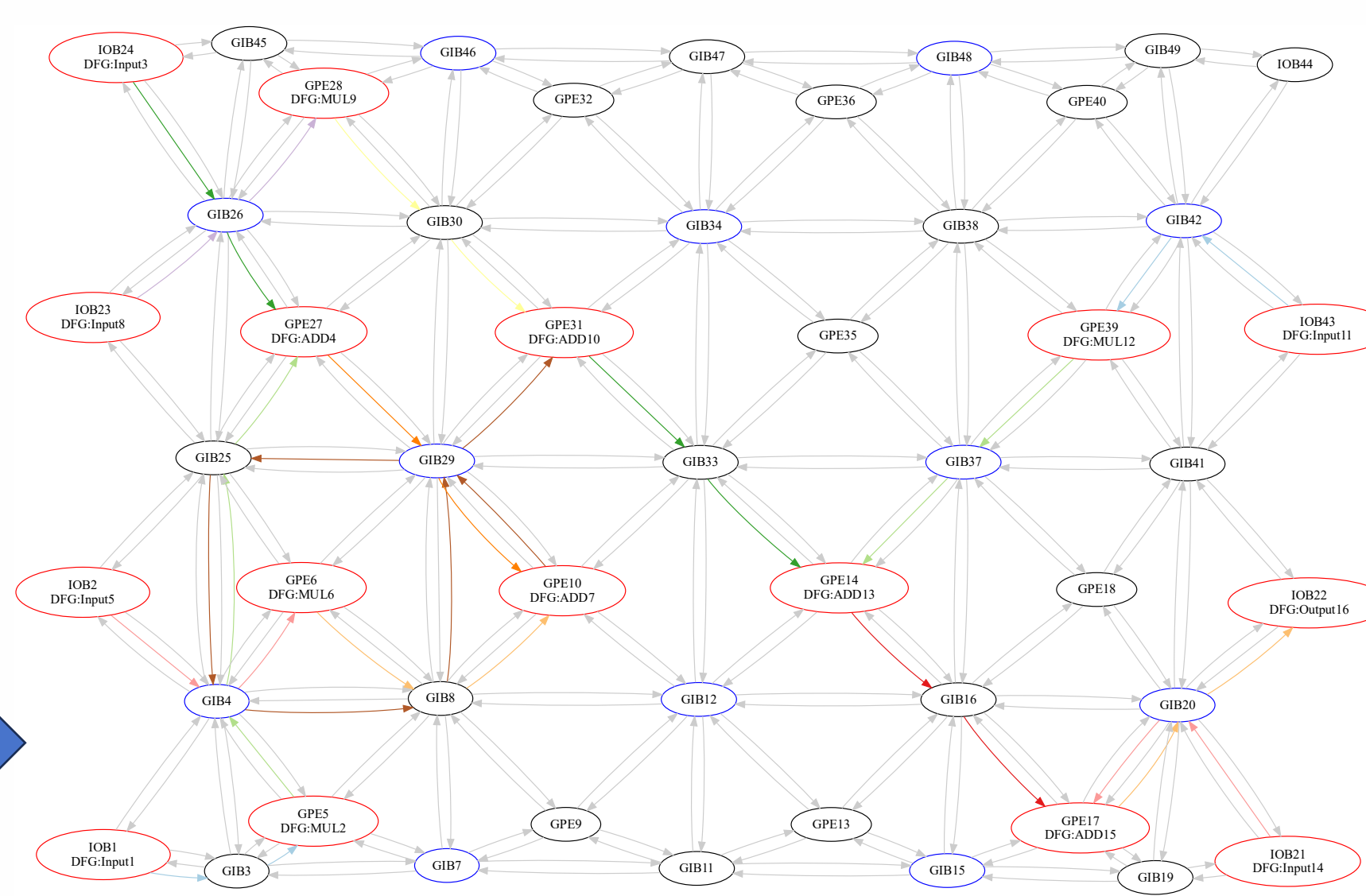
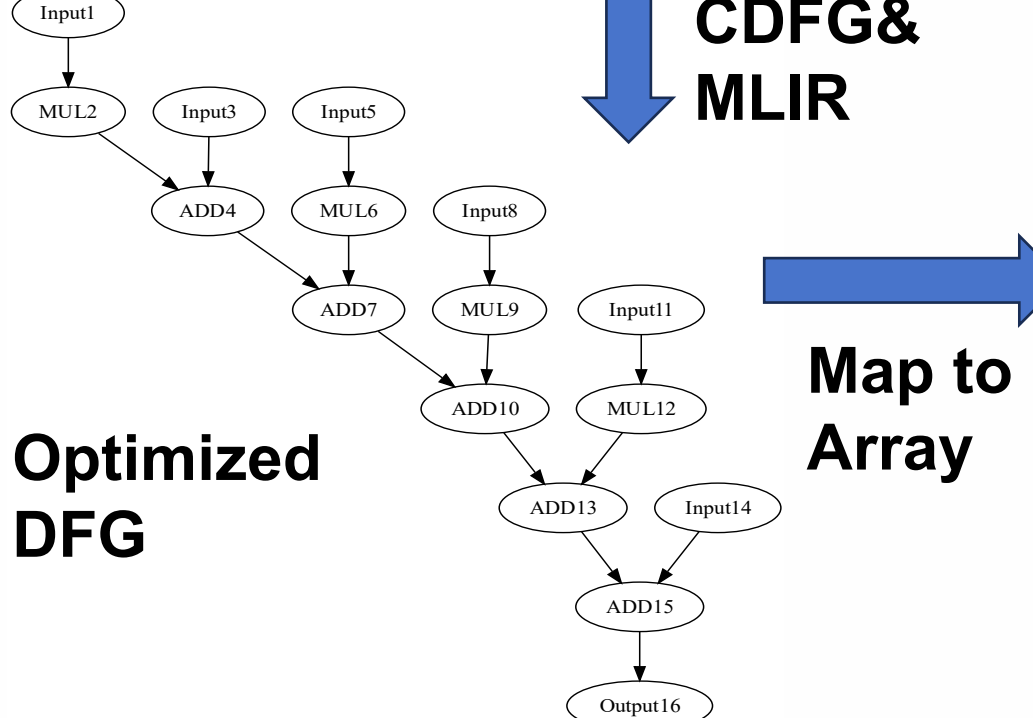
EXAMPLE: Convolution in Sobel Edge Detection

```

for y in range(1, height-1):
    for x in range(1, width-1):
        conv[(y-1)*(width-2) + (x-1)] = \
            gray[(y-1)*width + x-1]*(-1) + \
            gray[(y-1)*width + x+1]*(1) + \
            gray[y*width + x-1]*(-2) + \
            gray[y*width + x+1]*(2) + \
            gray[(y+1)*width + x-1]*(-1) + \
            gray[(y+1)*width + x+1]*(1)

```

Code to CDFG& MLIR



REFERENCES

- [1] J. Lou, Q. Zhu, Y. Dai et al., "Adora compiler: End-to-end optimization for high-efficiency dataflow acceleration and task pipelining on cgras," in 2025 62th ACM/IEEE Design Automation Conference (DAC), 2025, pp. 1377-1384.
- [2] Y. Qiu, Y. Mao, X. Gao et al., "FDRA: A framework for dynamically reconfigurable accelerator supporting multi-level parallelism," ACM Transactions on Reconfigurable Technology and Systems, vol. 17, no. 1, pp. 1-26, Jan. 2024.
- [3] Y. Dai, X. Gao, Y. Qiu et al., "COFFA: A co-design framework for fused-grained reconfigurable architecture towards efficient irregular loop handling," IEEE Transactions on Computers, vol. 74, no. 9, pp. 3099-3113, 2025.
- [4] A. Karpathy et al., "llama2.c," <https://github.com/karpathy/llama2.c>.

ACKNOWLEDGEMENT

Jiahang Lou and Jianrong Zhang are co-first authors. This work was supported by the National Science and Technology Major Project under Grant 2021ZD0114701.

復旦大學
FUDAN UNIVERSITY

ARC LAB



FPT25 Workshop



Jiahang's Homepage



Jianrong's Homepage



Yuan's Homepage

